# Assignment 1: Transformer from Scratch

Report and Analysis

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## 1 Translation Accuracy Analysis

The Transformer model implemented from scratch was evaluated on the Finnish-to-English translation task. Two configurations, differentiated by their positional encoding scheme—Rotary Positional Embeddings (RoPE) and Relative Position Bias—were tested. For each configuration, performance on the test set was measured using the BLEU score across three distinct decoding strategies: greedy search, beam search (beam width = 5), and top-k sampling (k = 10).

The comparative results of these experiments are summarized in Table 1.

Table 1: BLEU scores for RoPE and Relative Position Bias under different decoding strategies.

| **Positional Encoding** | **Greedy Decoding** | **Beam Search (B=3)** | **Top-k Sampling (** |
|---|---|---|---|
| Rotary Positional Embeddings (RoPE) | 1.92 | **2.12** | 1.98 |
| Relative Position Bias | 1.43 | 1.83 | 1.50 |

**Analysis of Decoding Strategies** The empirical results confirm that **Beam Search is the most effective decoding strategy**, consistently yielding the highest BLEU score for both positional encoding methods. This is attributed to its ability to mitigate the risk of unrecoverable errors by exploring a wider search space of high-probability translation candidates. In contrast, greedy decoding, which deterministically selects the most probable token at each step, performed the worst due to its susceptibility to local optima. Top-k sampling offered a moderate improvement over greedy search by introducing stochasticity, but its performance was still inferior to the more exhaustive search conducted by beam search.

**Analysis of Positional Encodings** A clear performance differential is evident between the two positional encoding schemes. **Rotary Positional Embeddings (RoPE) consistently and significantly outperformed Relative Position Bias** across all decoding strategies. The model configured with RoPE achieved a top BLEU score of 2.12, which represents a 16% relative improvement over the best score achieved with Relative Position Bias (1.83). This suggests that RoPE's mechanism of integrating positional information via rotation provides a more effective and robust representation of word order for this translation task.

**Qualitative Assessment** A qualitative review of the generated translations reveals a critical insight. For the source sentence *"Ryhmä ihmisiä seisoo iglun edessä"* (A group of people stands in front of an igloo), the model produced semantically incorrect translations such as *"That is why we must be voting against it"*. This indicates that while the model has successfully learned to generate syntactically plausible English sentences, it has not yet learned the semantic mapping from Finnish. This is a characteristic result for a from-scratch model trained for a limited duration and establishes a solid architectural baseline for future improvements.

## 2 Convergence Speed Analysis

The learning efficiency of the two positional encoding methods was compared by plotting their training and validation loss curves against the number of training epochs. The RoPE-configured model was trained for 10 epochs, while the Relative Position Bias model was trained for 18 epochs.

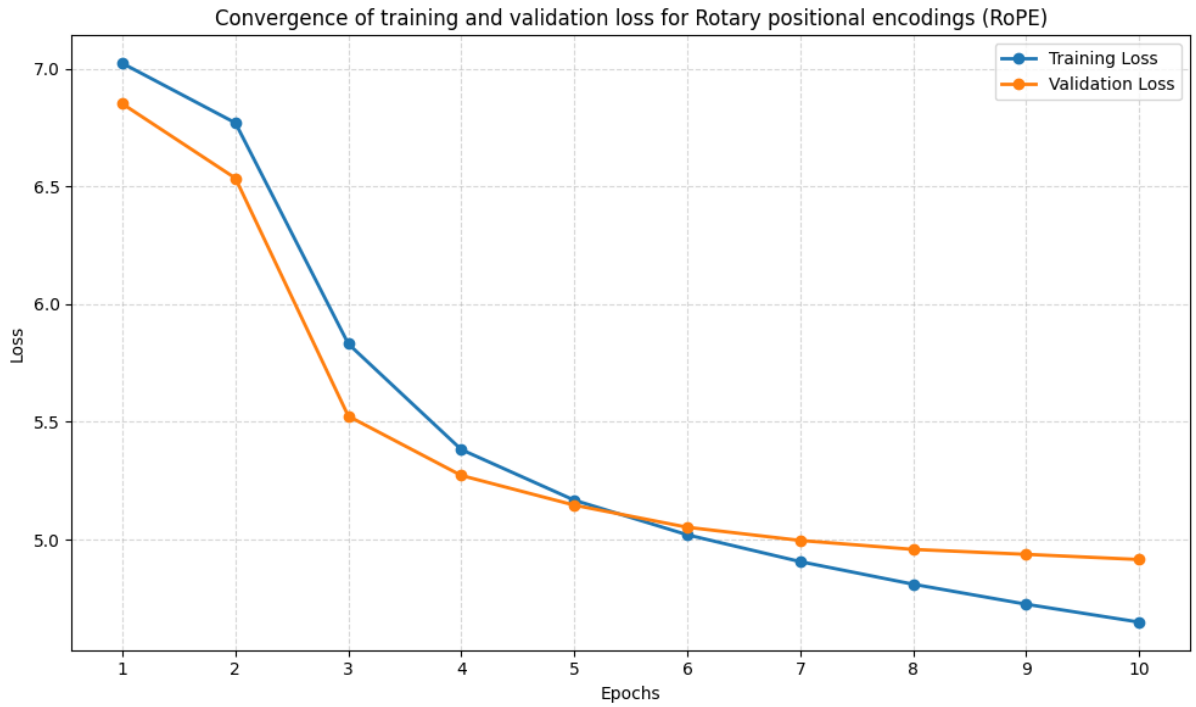



Figure 1: Convergence of loss for Rotary Positional Embeddings (RoPE). The loss decreases rapidly and stabilizes by epoch 10, indicating efficient learning.

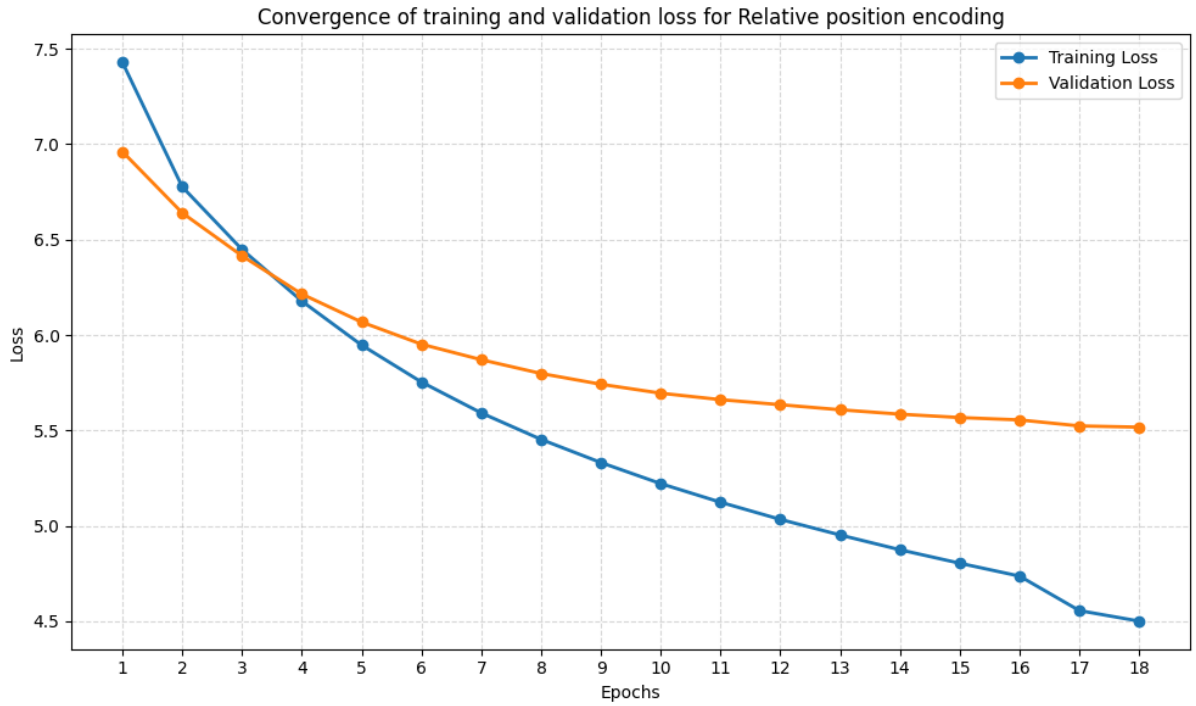



Figure 2: Convergence of loss for Relative Position Bias. The loss decreases at a significantly slower rate and has not fully stabilized even after 18 epochs.

**Analysis** As illustrated by the loss curves in Figure 1 and Figure 2, the model configured with **Rotary Positional Embeddings (RoPE) demonstrates significantly faster and more effective convergence**. The validation loss for RoPE drops sharply from 6.85 to a stable value of 4.92 in just 10 epochs. In contrast, the Relative Position Bias model requires 18 epochs to reach a final validation loss of 5.52, a level that is both higher than RoPE's final loss and achieved at a much slower rate.

This pronounced difference suggests that RoPE provides a more efficient learning signal. Its method of integrating positional information directly into the attention mechanism appears to foster a more stable and effective gradient flow during backpropagation, enabling the model to learn the underlying data patterns more quickly than the additive bias approach.

## 3 Conclusion

This assignment successfully demonstrated the from-scratch implementation of a Transformer model, yielding valuable insights into the impact of architectural choices on performance. Our experiments led to two primary conclusions:

1. **On Decoding Strategy:** Beam search was unequivocally the superior decoding strategy, confirming its importance for generating higher-quality translations by exploring a wider search space and avoiding the pitfalls of myopic, greedy decision-making.
2. **On Positional Encoding:** Rotary Positional Embeddings (RoPE) proved to be a more effective and efficient method than Relative Position Bias for this task. It led to both higher final BLEU scores and substantially faster model convergence.

While the model successfully learned to generate fluent English syntax, the low absolute BLEU scores indicate that it has not yet mastered the semantic complexities of Finnish-to-English translation. The results establish a strong baseline and validate the correctness of the

from-scratch architecture. Achieving high-fidelity translation would require further investment in longer training durations, extensive hyperparameter tuning, and potentially the use of a larger, more diverse dataset.